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multiple points of input, and better durability. What's more, capacitive screens allow the device to have a glass or some other rigid protection on top of the touch screen, thus improving the device's ergonomics.

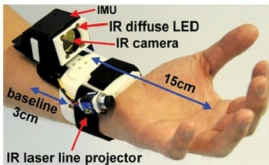
Touch-sensing controllers, such as the CapSense from Cypress, have specific technical improvements like water tolerant capacitive touch buttons, which eliminates false touches under moist conditions. "They also allow capacitive proximity sensing for up to 30cm, which is instrumental in enabling features such as 'wake-on approach,' and can even replace the IR sensors mentioned in the previous section," explains Anbarasu Samiappan, product apps manager, Cypress Semiconductor.

What's coming up in the future

I am pretty sure we will see something like Tom Cruise's sorcery in 'Minority Report' within a couple of decades. But until that happens, the field is taking great strides forward.

Researchers have been working on new kinds of hardware to compute gestures, including FPGAs. "Apart from basic components to capture the images, an FPGA-based gesture recognition system handling large computational complexity of optical flow to perform parallel processing is a low-cost solution. Using an FPGA system, gesture recognition can be implemented at 30 frames per second, and the system software can subsequently schedule all tasks during processing," explains P. Chow Reddy, manager (R&D), Power Division, ICOMM Tele Pvt Ltd.

Apart from the focus in consumer electronics, gesture recognition seems to have matured enough to be noticed and implemented in industrial settings. The BMW Group has demonstrated a gesture-driven system that allows a quality assurance worker to "...examine and



Microsoft's Digits prototype is part of an effort to bring gesture-based control in a mobile device. This wrist-worn sensor uses IR laser, LEDs and camera to recognise the fingers movement (Image courtesy: Microsoft)

document flaws in a component simply by pointing," Ford had earlier announced its own gesture-based ambitions in the form of 'a complete virtual factory.' Industry aside, even Microsoft Office 2013 has gesture recognition built into it.

At Electronics Rocks'13 conference, one of our speakers Hemant Surale showcased a cool smart glove product that he had designed. Hemant says that the present scenario in gesture recognition seems that development in this area is at its peak. "Most of the devices comes with their own proprietary SDKs for application developers. In addition there are many emerging startups developing effective way to interact with day to day equipments. But, Internet of Things will present new challenges with respect to integration of such systems into day to day life of users." Hemant is also an active member of the IoTBLR community. IoTBLR is a community based in Bangalore, which is excited about connecting disparate devices to one another via the web to create Internet of Things products. Any engineer can find them on facebook at [facebook.com/groups/IoTBLR](https://www.facebook.com/groups/IoTBLR)

The field of gesture recognition seems to be gearing up for quite a future, and design houses working on this technology better get ready for the competition that comes with any high-profile industry. ●

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